ESKWELA Software Testing & Quality Assurance (QA) Suite

Project: ESKWELA Android Mobile Application | **QA Lead:** Software Testing & QA Engineer

Code Repository Link: https://github.com/jameshofril06-jpg/SkwelaApp

Total Executed Test Cases: 80 Test Cases (50 Functional, 18 Integration/API, 12 UAT Scripts)

1. Executive Test Plan & QA Strategy

This document outlines the software quality assurance test suite for **ESKWELA**, containing 80 documented test cases, integration API test scripts, User Acceptance Testing (UAT) scenarios, and an initial bug tracking resolution matrix. Testing verifies functional correctness across Student and Administrator workflows.

2. Functional Testing Suite (50 Test Cases)

Part A: Authentication, Account Security & User Setup (TC-01 to TC-25)

TC ID	Module	Test Condition / Step Description	Expected Verification Result	Status
TC-01	Registration	Register with valid student credentials and password	Account created, redirected to Profile Setup	**PASSED**
TC-02	Registration	Register with short password (<6 chars)	Displays validation error toast	**PASSED**
TC-03	Registration	Register with duplicate existing email	Firebase Auth rejects duplicate registration	**PASSED**
TC-04	Login	Enter valid registered student credentials	Authenticated successfully, opens Home Dashboard	**PASSED**
TC-05	Login	Enter invalid password for registered account	Displays invalid credentials error toast	**PASSED**
TC-06	Login	Tap 'Forgot Password' link and submit email	Password reset email dispatched via Firebase	**PASSED**
TC-07	Profile Setup	Enter Full Name, School, Course, Year, Section	Profile stored in Firestore document	**PASSED**
TC-08	Profile Setup	Leave Full Name blank during profile setup	Blocks submission with 'Full Name required' error	**PASSED**
TC-09	Account Guard	Attempt login with suspended student status	Blocked with restriction notice, session terminated	**PASSED**
TC-10	Account Guard	Attempt login with deactivated account status	Blocked with account deactivated alert	**PASSED**
TC-11	Account Guard	Attempt login with archived account status	Blocked with account archived alert	**PASSED**
TC-12	Admin Auth	Login using administrator email credentials	AccountGuard redirects directly to Admin Center	**PASSED**
TC-13	Admin Isolation	Admin attempts opening student schedule tab	Access restricted, redirects to Admin Center	**PASSED**
TC-14	Admin Isolation	Admin attempts opening student tasks tab	Access restricted, redirects to Admin Center	**PASSED**
TC-15	Admin Isolation	Admin attempts opening student study space	Access restricted, redirects to Admin Center	**PASSED**
TC-16	Logout	Tap Logout button in Profile settings	Displays confirmation dialog (Yes/Cancel)	**PASSED**
TC-17	Logout	Confirm 'Yes, Logout' in confirmation popup	Revokes session, clears backstack to Landing screen	**PASSED**
TC-18	Logout	Tap 'Cancel' in Logout confirmation popup	Dialog dismisses, user remains on Profile screen	**PASSED**
TC-19	Dark Mode	Toggle Dark Theme switch in Profile settings	App appearance instantly flips to dark theme	**PASSED**
TC-20	Dark Mode	Edit profile in Dark Mode	Text inputs and labels render in high contrast	**PASSED**
TC-21	OTA Check	Tap 'Check for update (OTA)' when on v1.1.5	Displays 'You're Up to Date!' dialog	**PASSED**

TC-22	OTA Download	Tap 'Check for update (OTA)' when on older build	Displays release notes and live progress bar	**PASSED**
TC-23	About App	Tap 'About this app' in Profile settings	Displays version info and complete changelog	**PASSED**
TC-24	Online Sync	Toggle Online Auto-Sync off/on	Enables/disables Firestore network connection	**PASSED**
TC-25	Profile View	Open Profile tab as student	Displays avatar initials, name, school, course, year	**PASSED**

Part B: Study Load OCR, Timetable, Tasks, AI Quizzes & Admin (TC-26 to TC-50)

TC ID	Module	Test Condition / Step Description	Expected Verification Result	Status
TC-26	OCR Timetable	Tap Camera scan button in schedule setup	Launches camera capture, reads study load text	PASSED
TC-27	OCR Timetable	Tap Gallery import button in schedule setup	Opens photo picker, extracts timetable OCR text	PASSED
TC-28	OCR Timetable	Scan sample study load document image	Auto-plots 6 courses across weekdays	PASSED
TC-29	Day View	Tap Monday date tab on schedule calendar	Displays ONLY Monday classes	PASSED
TC-30	Day View	Tap Tuesday date tab on schedule calendar	Displays ONLY Tuesday classes	PASSED
TC-31	Week View	Tap 'Week' toggle tab on schedule calendar	Displays all enrolled classes for current week	PASSED
TC-32	Month View	Tap 'Month' toggle tab on schedule calendar	Displays all enrolled classes for current month	PASSED
TC-33	Add Class	Add manual class with valid times and room	Class saved to Firestore, updates timetable	PASSED
TC-34	Time Conflict	Add class overlapping existing class time	Displays ■■ Conflict warning with override option	PASSED
TC-35	Delete Class	Long press class card and type matching Class Code	Validates code and deletes class from schedule	PASSED
TC-36	Delete Class	Long press class card and type WRONG Class Code	Blocks deletion with validation error toast	PASSED
TC-37	Task Creation	Create new task with title, date, priority, course	Task saved, rendered in task list with badge	PASSED
TC-38	Task Complete	Tap checkbox on task item	Toggles task completion status and updates subtitle	PASSED
TC-39	Task Filter	Filter tasks by 'All', 'Pending', or 'Completed'	Filters task list accurately	PASSED
TC-40	Task Drop	Tap delete task button and enter drop reason	Prompts confirmation and records drop reason	PASSED
TC-41	AI Quiz Pre-fill	Open AI Quiz activity as student	Pre-fills topic with student's enrolled course	PASSED
TC-42	AI Quiz Picker	Tap 'Study Load' button in AI Quiz	Displays course selection dialog of active courses	PASSED
TC-43	AI Quiz Count	Select 5, 10, or 15 questions in AI Quiz	Updates selected question count segment	PASSED
TC-44	AI Quiz Runner	Submit answers through AI Practice Quiz	Tracks score, displays answer explanations	PASSED
TC-45	Leaderboard	Complete AI Quiz with high percentage	Saves score, updates live Leaderboard rankings	PASSED
TC-46	Focus Timer	Select focus timer preset (2m, 5m, 10m, 15m, 25m, 45m, 60m)	Updates timer display duration	PASSED
TC-47	Focus Timer	Start, pause, and reset focus countdown	Controls timer execution state	PASSED
TC-48	Admin Overview	Open Admin Control Center as administrator	Displays Total Users and System Classes stats	PASSED
TC-49	Admin Moderation	Tap student item in Admin Control Center	Opens slide-up pane showing remaining suspension time	PASSED
TC-50	Admin Broadcast	Type and post official announcement in Admin Center	Broadcasts announcement to all student dashboards	PASSED

3. Integration & REST API Test Suite (18 Test Cases)

Integration & REST API Testing (INT-01 to INT-18)

TC ID	Module	Test Condition / Step Description	Expected Verification Result	Status
INT-01	Firebase Auth API	POST /v1/accounts:signUp with email/pass	Returns HTTP 200 OK with idToken and refreshToken	PASSED
INT-02	Firebase Auth API	POST /v1/accounts:signInWithPassword	Returns HTTP 200 OK with authenticated user UID	PASSED
INT-03	Firestore Users API	GET /documents/users/{userId}	Returns user profile document JSON fields	PASSED
INT-04	Firestore Users API	PATCH /documents/users/{userId} status='suspended'	Updates user status and suspensionDuration in Firestore	PASSED
INT-05	Firestore Users API	PATCH /documents/users/{userId} status='archived'	Updates user status to archived	PASSED
INT-06	Firestore Schedules	POST /documents:runQuery for schedules dateString	Returns JSON array of enrolled classes for date	PASSED
INT-07	Firestore Schedules	POST /documents/schedules (Add Class)	Creates new class schedule document in Firestore	PASSED
INT-08	Firestore Schedules	DELETE /documents/schedules/{scheduleId}	Removes schedule document from Firestore	PASSED
INT-09	Firestore Tasks API	POST /documents:runQuery for tasks completed==false	Returns pending task items for user	PASSED
INT-10	Firestore Tasks API	POST /documents/tasks (Add Task)	Creates task document with priority and course link	PASSED
INT-11	Firestore Tasks API	PATCH /documents/tasks/{taskId} completed=true	Updates task completion flag in Firestore	PASSED
INT-12	Firestore Tasks API	DELETE /documents/tasks/{taskId}	Deletes task document from Firestore	PASSED
INT-13	Firestore Quiz API	POST /documents/ai_quizzes (Save Quiz Attempt)	Saves score, topic, percentage, timestamp	PASSED
INT-14	Firestore Quiz API	POST /documents:runQuery ORDER BY percentage DESC	Returns top 5 high scores for Leaderboard	PASSED
INT-15	Firestore Broadcast	POST /documents/announcements (Post Broadcast)	Creates announcement document with postedAt timestamp	PASSED
INT-16	Firestore Broadcast	POST /documents:runQuery ORDER BY postedAt DESC	Returns latest announcement for Home banner	PASSED
INT-17	Firestore Focus API	POST /documents/focus_sessions (Record Session)	Stores completed focus session duration	PASSED
INT-18	GitHub Releases API	GET api.github.com/repos/.../releases	Returns HTTP 200 OK JSON array of release tags and assets	PASSED

4. User Acceptance Testing (UAT) Scripts (12 Test Cases)

User Acceptance Testing (UAT) Scenario Scripts (UAT-01 to UAT-12)

TC ID	Module	Test Condition / Step Description	Expected Verification Result	Status
UAT-01	Student Onboarding	Register → Complete Profile Setup → Scan Timetable	Student successfully completes onboarding flow	PASSED
UAT-02	Study Load Scanning	Scan study load document via Camera → Verify timetable	All 6 enrolled courses automatically plotted across days	PASSED
UAT-03	Schedule Calendar	Switch between Day, Week, and Month views on timetable	Calendar displays date-specific classes accurately	PASSED
UAT-04	Conflict Prevention	Try adding a class that overlaps an existing class time	Warning popup alerts user, allows override or cancel	PASSED
UAT-05	Class Deletion	Long press class card → Type matching class code	Class code validated and removed from schedule	PASSED
UAT-06	Task Management	Create task → Set HIGH priority → Toggle complete → Drop task	Task lifecycle executed with drop reason tracking	PASSED

UAT-07	AI Practice Quiz	Select enrolled course → Take 10 Qs → Review answers	Quiz completed, explanations shown, score saved	**PASSED**
UAT-08	Leaderboard Rank	Achieve 100% score on AI Quiz → View Leaderboard	User ranking rendered with medal badge on Leaderboard	**PASSED**
UAT-09	Pomodoro Focus	Select 25m preset → Start focus timer → Complete session	Timer completes and records session in history	**PASSED**
UAT-10	Admin Moderation	Admin logs in → Search student → Apply 24h suspension	Student account suspended, remaining time tracked	**PASSED**
UAT-11	Admin Broadcast	Admin posts official announcement → Student views banner	Announcement appears live on student Home dashboard	**PASSED**
UAT-12	OTA Update Flow	User checks for OTA update → Progress bar → Install	Update detected, progress tracked, installer opened	**PASSED**

5. Initial Bug Tracking Log & Resolution Matrix

Bug ID	Description / Defect	Severity	Root Cause Analysis	Resolution / Fix Applied	Status
BUG-01	Scanned study load did not reflect on specific day tabs	HIGH	Schedules were only seeded for specific dates instead of full weekdays.	Updated ScheduleActivity to plot study load across all weekdays.	RESOLVED
BUG-02	OTA update download failed with HTTP 404	HIGH	DownloadManager failed on GitHub's /releases/latest/ 302 redirect.	Switched to GitHub /releases API index 0 and added cache-busting timestamp.	RESOLVED
BUG-03	Text unreadable when editing profile in Dark Mode	MEDIUM	Profile setup layout contained hardcoded dark hex text colors.	Replaced hardcoded hex colors with dynamic @color/text_primary tokens.	RESOLVED
BUG-04	Time schedule overlap allowed duplicate class booking	MEDIUM	AddClassActivity lacked start/end time overlap validation.	Implemented parseTimeToMinutes() and overlap check with warning popup.	RESOLVED
BUG-05	Task deletion occurred accidentally on single tap	LOW	No confirmation dialog before removing task records.	Added AlertDialog requiring confirmation and drop reason prompt.	RESOLVED
BUG-06	Admin user could access student schedule/tasks tabs	HIGH	Role access guard lacked strict navigation redirection for admin users.	Implemented AccountGuard.redirectIfAdmin() routing admins exclusively to Admin Center.	RESOLVED
BUG-07	OTA update check returned 'You are up to date' for pre-releases	MEDIUM	GitHub /releases/latest API ignores pre-release tags.	Updated UpdateHelper to query /releases list index 0 including pre-releases.	RESOLVED
BUG-08	Class deletion had no accidental removal protection	MEDIUM	Long press deleted class without code verification.	Added professional validation dialog requiring typing matching Class Code.	RESOLVED
BUG-09	Home Dashboard showed classes from all days	MEDIUM	Date query lacked strict in-memory date prefix filtering.	Enforced strict startsWith(todayDateString) filtering in DashboardActivity.	RESOLVED
BUG-10	Stale APK installed during OTA update from Downloads	MEDIUM	DownloadManager did not clear previous cached eskwela-latest.apk file.	Added file deletion cleanup and dynamic URL timestamp parameter ?t=time.	RESOLVED